THE COMPLETE GUIDE TO

150 REAL-TIME AI INTERVIEW QUESTIONS

REAL QUESTIONS. DEEP FOLLOW-UPS.

AI ENGINEER | GENERATIVE AI ENGINEER | LLM ENGINEER
ML ENGINEER | AI ARCHITECT



PYTHON FOR AI
Q1 – Q25
Core Python, OOP, Libraries, Data Structures & More

MACHINE LEARNING
Q26 – Q50
Algorithms, Evaluation, Feature Engineering & More

DEEP LEARNING
Q51 – Q75
Neural Networks, CNN, RNN, Training, Optimization & More

GENERATIVE AI & LLMS
Q76 – Q100
LLM Architecture, Prompt Engineering, Fine-Tuning, RLHF, LoRA & More

RAG & VECTOR DATABASE
Q101 – Q120
RAG Architecture, Chunking, Embeddings, Vector DBs & More

AGENTIC AI & MULTI-AGENT SYSTEMS
Q121 – Q135
AI Agents, Tools, Memory, Planning, Multi-Agent Collaboration & More

SYSTEM DESIGN & LLMOPS
Q136 – Q150
AI System Design, Deployment, Monitoring, Scaling, Security & More

FAANG LEVEL PREPARATION | PRACTICE REAL INTERVIEW QUESTIONS | CONCEPTS MADE EASY | PREPARE SMART CRACK BIG

MASTER AI. CRACK INTERVIEWS. BUILD THE FUTURE.

150 Real-Time AI Interview Questions (2026-2027 Edition)

AI Engineer | Generative AI Engineer | LLM Engineer | ML Engineer | AI Architect

Real Interview Questions + Follow-up Questions + Expected Discussion Points

SECTION 1: Python & AI Coding Interview Questions

Q1. Implement Cosine Similarity for AI Embeddings

Interviewer Asked:

Can you write Python code to calculate cosine similarity between two embedding vectors without using any machine learning library?

Follow-up Questions Interviewer Can Ask:

1. Explain why cosine similarity is widely used in semantic search and RAG systems.
2. What is the difference between cosine similarity and Euclidean distance?
3. If two vectors have the same direction but different magnitude, what will cosine similarity return?
4. How does a vector database perform similarity search internally?
5. How would you optimize similarity search when you have billions of vectors?

Q2. Build an Embedding Pipeline for Millions of Documents

Interviewer Asked:

Your company has 10 million documents and wants to build an AI search system. How would you design the embedding generation pipeline?

Follow-up Questions Interviewer Can Ask:

1. How will you split large documents before generating embeddings?
2. How will you decide the chunk size and overlap?
3. Which embedding model would you choose and why?
4. How will you reduce embedding generation cost?
5. How will you update embeddings when a document changes?
6. How would you handle duplicate documents?

Q3. Build a Neural Network From Scratch

Interviewer Asked:

Can you implement a simple neural network using Python and NumPy without using TensorFlow or PyTorch?

Follow-up Questions Interviewer Can Ask:

1. Explain the complete flow of forward propagation.
2. How does backpropagation calculate gradients?
3. What happens if the learning rate is too high?
4. Why do we need activation functions?
5. How does PyTorch perform automatic differentiation?
6. What is a computational graph?

Q4. Optimize a Slow Machine Learning Pipeline

Interviewer Asked:

Your AI model training pipeline takes 12 hours. How would you reduce the training time?

Follow-up Questions Interviewer Can Ask:

1. How will you identify the bottleneck?
2. Would you optimize the code, hardware, or model architecture first?
3. How does GPU acceleration improve training?
4. How would you optimize data loading?
5. What techniques can reduce model training cost?

Q5. Design a Production ML Data Pipeline

Interviewer Asked:

Design a scalable data preprocessing pipeline for a machine learning system handling millions of records.

Follow-up Questions Interviewer Can Ask:

1. How will you handle missing values?
2. How will you detect bad quality data?
3. How do you prevent data leakage?
4. How will you validate incoming production data?
5. How would you monitor this pipeline?

Q6. Implement a Machine Learning Evaluation System

Interviewer Asked:

Write a function to calculate precision, recall, and F1-score without using sklearn.

Follow-up Questions Interviewer Can Ask:

1. Why is accuracy not enough for many AI applications?
2. For a fraud detection system, would you optimize precision or recall?
3. How do you evaluate models on imbalanced datasets?
4. Explain confusion matrix.
5. How do you select evaluation metrics for an AI product?

Q7. Process Large AI Datasets

Interviewer Asked:

You need to process a 1TB dataset but your server has only 16GB RAM. How will you solve this problem?

Follow-up Questions Interviewer Can Ask:

1. Why would you use generators instead of loading everything into memory?"
2. When would you use distributed computing?
3. How does Spark help in large-scale AI processing?
4. How would you optimize memory usage?

Q8. Deploy a Machine Learning Model as an API

Interviewer Asked:

Your ML model is ready. How will you deploy it as a production API used by thousands of customers?

Follow-up Questions Interviewer Can Ask:

1. Which framework would you use for deployment?
2. How will you handle high traffic?
3. How will you reduce API latency?
4. How will you monitor model performance?
5. How will you roll back a bad model deployment?

Q9. Build a Real-Time AI Recommendation System

Interviewer Asked:

Design a recommendation system similar to Netflix or Amazon.

Follow-up Questions Interviewer Can Ask:

1. How will you collect user behavior data?
2. Which recommendation algorithm will you choose?
3. How will you handle cold-start problems?
4. How will you scale recommendations for millions of users?
5. How will you evaluate recommendation quality?

Q10. Build an AI Chatbot Backend

Interviewer Asked:

Design the backend architecture for an AI chatbot application.

Follow-up Questions Interviewer Can Ask:

1. How will you maintain conversation memory?
2. How will you reduce LLM API cost?
3. How will you handle millions of conversations?
4. How will you prevent hallucinations?
5. How will you monitor chatbot quality?

Q11. Implement a Text Similarity Search System

Interviewer Asked:

How would you build a text similarity search system where users can find semantically similar documents?

Follow-up Questions Interviewer Can Ask:

1. Why is keyword search not enough for modern AI search systems?
2. Explain how embeddings help in semantic search.
3. Which embedding model would you choose and why?
4. How would you store and search millions of embeddings?
5. How would you improve search relevance?

Q12. Build a Document Chunking Strategy for LLM Applications

Interviewer Asked:

You are building a RAG application. How would you split large documents into chunks before sending them to an LLM?

Follow-up Questions Interviewer Can Ask:

1. What happens if chunk size is too small?
2. What happens if chunk size is too large?
3. Explain fixed chunking vs semantic chunking.
4. How do you decide chunk overlap?
5. How would you handle tables and images inside documents?

Q13. Optimize LLM Application Latency

Interviewer Asked:

Your GenAI application response time is 15 seconds. How would you reduce latency?

Follow-up Questions Interviewer Can Ask:

1. Which components contribute to LLM latency?
2. How does token count affect response time?
3. Explain streaming responses.
4. How does KV cache improve inference speed?
5. How would you optimize cost and latency together?

Q14. Design a Model Monitoring System

Interviewer Asked:

After deploying an AI model, how will you monitor whether it is working correctly?

Follow-up Questions Interviewer Can Ask:

1. What metrics will you monitor?
2. How will you detect model drift?
3. How will you detect data drift?
4. How will you identify model failures?
5. What alerts would you configure

Q15. Build a Machine Learning Training Pipeline

Interviewer Asked:

Design an automated ML training pipeline from data collection to model deployment.

Follow-up Questions Interviewer Can Ask:

1. How will you automate model retraining?
2. How will you manage different model versions?
3. How will you track experiments?
4. How will you rollback a bad model?
5. Which MLOps tools would you use?

Q16. Handle Missing Data in AI Systems

Interviewer Asked:

Your training dataset contains 30% missing values. How would you handle it?

Follow-up Questions Interviewer Can Ask:

1. When would you remove missing data?
2. When would you impute missing values?
3. Which imputation techniques have you used?
4. How can missing values impact model performance?
5. How do you handle missing data in production?

Q17. Detect and Remove Data Bias

Interviewer Asked:

Your AI model produces biased predictions. How would you identify and fix the issue?

Follow-up Questions Interviewer Can Ask:

1. What causes bias in AI models?
2. How do you measure fairness?
3. How can training data introduce bias?
4. How do you create responsible AI systems?
5. Explain fairness vs accuracy tradeoff.

Q18. Design an AI Recommendation Engine

Interviewer Asked:

Design a recommendation system for an e-commerce platform with millions of users.

Follow-up Questions Interviewer Can Ask:

What data will you collect?
Collaborative filtering vs content-based filtering?
How will you solve the cold-start problem?
How will you handle real-time recommendations?
How will you measure recommendation quality?

Q19. Build a Fraud Detection AI System

Interviewer Asked:

Design an AI-based fraud detection system for a banking application.

Follow-up Questions Interviewer Can Ask:

Why is fraud detection an imbalanced classification problem?
Which metrics will you optimize?
How will you reduce false positives?
How will you detect new fraud patterns?
How will you deploy this system in real time?

Q20. Explain Model Deployment Architecture

Interviewer Asked:

Your AI model is trained successfully. Explain how you would deploy it into production.

Follow-up Questions Interviewer Can Ask:

Batch inference vs real-time inference?
How will you handle thousands of requests per second?
How will you manage GPU resources?
How will you monitor API failures?
How will you update the model without downtime?

Q21. Build an AI Data Validation System

Interviewer Asked:

How would you validate incoming data before it reaches your ML model?

Follow-up Questions Interviewer Can Ask:

What data quality checks would you implement?
How do you detect schema changes?
How do you handle corrupted data?
How do you automate validation?

Q22. Implement AI Model Version Control

Interviewer Asked:

How do you manage multiple versions of machine learning models in production?

Follow-up Questions Interviewer Can Ask:

How do you compare model versions?
What is A/B testing for AI models?
How do you rollback a model?
How do you track model experiments?

Q23. Design a Real-Time Prediction System

Interviewer Asked:

Design a system that can generate AI predictions within milliseconds.

Follow-up Questions Interviewer Can Ask:

How will you reduce inference latency?
Where will you use caching?
How will you scale the system?
How will you handle traffic spikes?

Q24. Explain Feature Engineering for AI Models

Interviewer Asked:

How do you decide which features to create for a machine learning model?

Follow-up Questions Interviewer Can Ask:

How do you identify useful features?
Can too many features hurt performance?
Explain feature selection techniques.
How do you handle changing features in production?

Q25. Design an End-to-End AI Solution

Interviewer Asked:

You are given a business problem. Explain how you would convert it into a complete AI solution.

Follow-up Questions Interviewer Can Ask:

How do you define the AI problem?
How do you collect training data?
How do you select the model?
How do you deploy and monitor it?
How do you measure business impact?

SECTION 2: Machine Learning Interview Questions

Q26. Explain Your Machine Learning Project End-to-End

Interviewer Asked:

Explain a machine learning project you have worked on from data collection to deployment.

Follow-up Questions Interviewer Can Ask:

What business problem were you solving?
How did you collect and prepare the data?
Why did you choose that specific ML algorithm?
What challenges did you face during development?
How did you evaluate model performance?
How did you deploy the model into production?

Q27. Select the Right Machine Learning Algorithm

Interviewer Asked:

Given a business problem, how do you decide which machine learning algorithm to use?

Follow-up Questions Interviewer Can Ask:

How do you decide between classification and regression?
When would you choose a tree-based model over a neural network?
How does dataset size influence algorithm selection?
How do accuracy requirements affect your choice?
Would you always choose the most complex model?

Q28. Debug a Model With High Accuracy but Poor Production Results

Interviewer Asked:

Your ML model achieves 98% accuracy during testing but performs poorly after deployment. How would you debug it?

Follow-up Questions Interviewer Can Ask:

Could this be caused by data leakage?
How would you check for data drift?
How would you compare training data with production data?
How would you identify whether the problem is the model or data?
What monitoring would you add?

Q29. Explain Bias and Variance Tradeoff

Interviewer Asked:

Explain the bias-variance tradeoff with a real-world example.

Follow-up Questions Interviewer Can Ask:

How do you identify high bias in a model?
How do you identify high variance?
How would you reduce overfitting?
How would you improve an underfitting model?
Can a model have both high bias and high variance?

Q30. Prevent Overfitting in Machine Learning Models

Interviewer Asked:

Your model performs very well on training data but poorly on unseen data. How will you fix it?

Follow-up Questions Interviewer Can Ask:

Explain regularization techniques.
Difference between L1 and L2 regularization?
How does dropout reduce overfitting?
How does increasing training data help?
When would you reduce model complexity?

Q31. Handle Imbalanced Datasets

Interviewer Asked:

You are building a fraud detection model where only 1% of transactions are fraudulent. How would you handle this imbalance?

Follow-up Questions Interviewer Can Ask:

Why is accuracy a bad metric here?
Explain precision vs recall for fraud detection.
When would you use SMOTE?
How do class weights work?
Which evaluation metric would you select?

Q32. Explain Supervised, Unsupervised, and Reinforcement Learning

Interviewer Asked:

Explain different types of machine learning with real-world examples.

Follow-up Questions Interviewer Can Ask:

When would you use supervised learning?
Give examples of unsupervised learning problems.
How does reinforcement learning learn from rewards?
Where is reinforcement learning used in industry?

Q33. Random Forest vs XGBoost

Interviewer Asked:

"Compare Random Forest and XGBoost. When would you use each?"

Follow-up Questions Interviewer Can Ask:

How does Random Forest reduce variance?
How does boosting improve weak learners?
Why is XGBoost popular in competitions?
Which model handles missing values better?
Which model would you use for tabular enterprise data?

Q34. Explain Gradient Boosting Algorithm

Interviewer Asked:

Explain how gradient boosting works internally.

Follow-up Questions Interviewer Can Ask:

Why are models trained sequentially in boosting?
How does each new tree improve previous predictions?
What is the role of the learning rate?
Difference between bagging and boosting?

Q35. Decision Tree Splitting

Interviewer Asked:

How does a decision tree decide the best feature for splitting data?

Follow-up Questions Interviewer Can Ask:

Explain entropy and information gain.
What is Gini impurity?
Why do decision trees overfit easily?
How do Random Forests solve this problem?

Q36. Feature Engineering Strategy

Interviewer Asked:

Explain your approach to feature engineering for a machine learning project.

Follow-up Questions Interviewer Can Ask:

How do you identify important features?
How do you create new features?
How do you handle categorical variables?
Can too many features reduce model performance?
How do you automate feature engineering?

Q37. Handle Missing Data

Interviewer Asked:

Your dataset contains missing values. How would you handle them before training?

Follow-up Questions Interviewer Can Ask:

When would you remove rows with missing values?
Explain different imputation methods.

How can missing values create bias?
How would you handle missing values during prediction?

Q38. Feature Selection Techniques

Interviewer Asked:

How do you select the most important features for your model?

Follow-up Questions Interviewer Can Ask:

Explain filter, wrapper, and embedded methods.
How does feature importance work?
Why does reducing features sometimes improve performance?
How does PCA help with feature reduction?

Q39. Precision, Recall, and F1 Score

Interviewer Asked:

Explain precision, recall, and F1-score. When would you optimize each metric?

Follow-up Questions Interviewer Can Ask:

For cancer detection, which metric is more important?
For spam detection, which metric matters more?
Explain confusion matrix.
Why can accuracy be misleading?

Q40. Evaluate Machine Learning Models

Interviewer Asked:

How do you evaluate whether a machine learning model is ready for production?

Follow-up Questions Interviewer Can Ask:

Which offline metrics do you check?
How do you perform production testing?
Explain A/B testing for ML models.
How do you measure business impact?

Q41. Model Performance Suddenly Drops in Production

Interviewer Asked:

Your machine learning model was performing well for six months, but suddenly the accuracy dropped. How would you investigate the issue?

Follow-up Questions Interviewer Can Ask:

How would you identify whether the issue is with data or the model?
Explain data drift and concept drift.
What monitoring metrics would you check first?
How would you compare production data with training data?
Would you retrain the model immediately? Why or why not?
How would you prevent this issue in future?

Q42. Explain Data Drift and Model Drift

Interviewer Asked:

What is data drift and model drift? Explain with real-world examples.

Follow-up Questions Interviewer Can Ask:

How do you detect data drift automatically?
What statistical techniques can be used for drift detection?
How does drift affect model performance?
How frequently should models be retrained?
How would you design a drift monitoring system?

Q43. Monitor a Machine Learning Model in Production

Interviewer Asked:

After deploying an ML model, what metrics would you monitor?

Follow-up Questions Interviewer Can Ask:

Which technical metrics would you track?
Which business metrics would you track?
How would you detect prediction failures?
How would you monitor latency?
How would you create alerts for model issues?

Q44. Hyperparameter Optimization

Interviewer Asked:

How do you tune hyperparameters for a machine learning model?

Follow-up Questions Interviewer Can Ask:

Explain Grid Search and Random Search.
Why is Random Search sometimes better than Grid Search?
What is Bayesian optimization?
How do you avoid overfitting during tuning?
Which parameters would you tune for XGBoost?

Q45. Grid Search vs Random Search

Interviewer Asked:

Compare Grid Search and Random Search for hyperparameter tuning.

Follow-up Questions Interviewer Can Ask:

Which method is faster and why?
When would Grid Search be preferred?
How does Bayesian optimization improve tuning?
How do you tune deep learning models differently?

Q46. Cross Validation

Interviewer Asked:

Explain cross-validation and why it is important in machine learning.

Follow-up Questions Interviewer Can Ask:

Why not train-test split only?
Explain K-fold cross-validation.
What happens if the dataset is time-series data?
When should you avoid random cross-validation?
How does cross-validation help prevent overfitting?

Q47. Reduce False Positives in Classification Systems

Interviewer Asked:

Your AI system generates too many false alarms. How would you reduce false positives?

Follow-up Questions Interviewer Can Ask:

Which metric would you optimize?
How does changing the classification threshold help?
Would you modify data or the model?
Explain precision-recall tradeoff.
Give an example where false positives are expensive.

Q48. Improve an Underperforming ML Model

Interviewer Asked:

Your machine learning model is not achieving the expected performance. What steps would you take?

Follow-up Questions Interviewer Can Ask:

Would you collect more data or change the model first?
How do you check data quality?
How do you analyze model errors?
How do you improve feature engineering?
When would you replace the algorithm completely?

Q49. Explain Ensemble Learning

Interviewer Asked:

Explain ensemble learning and why combining multiple models can improve performance.

Follow-up Questions Interviewer Can Ask:

Explain bagging and boosting.
How does Random Forest use ensemble learning?
How does XGBoost combine weak learners?

Can ensemble models overfit?
When would you avoid ensemble methods?

Q50. Design an End-to-End Machine Learning Solution

Interviewer Asked:

You are given a business problem. Explain how you would design an end-to-end machine learning solution.

Follow-up Questions Interviewer Can Ask:

How do you convert a business problem into an ML problem?
How do you collect and prepare data?
How do you select the right model?
How do you deploy the solution?
How do you monitor the model after deployment?
How do you calculate ROI from an AI solution?

SECTION 3: Deep Learning & Transformer Interview Questions

Q51. Explain Neural Network Architecture

Interviewer Asked:

Explain the architecture of a neural network and how data flows through different layers.

Follow-up Questions Interviewer Can Ask:

What happens inside an input layer, hidden layer, and output layer?
How does a neuron calculate its output?
Why do we need multiple hidden layers?
What happens if we increase the number of layers?
How do you decide the number of layers and neurons?

Q52. Explain Forward Propagation and Backpropagation

Interviewer Asked:

Explain how forward propagation and backpropagation work during neural network training.

Follow-up Questions Interviewer Can Ask:

How is the loss calculated?
How are gradients calculated?
How are weights updated?
Why is backpropagation computationally expensive?
How does automatic differentiation work in PyTorch?

Q53. Why Do Neural Networks Need Activation Functions?

Interviewer Asked:

Why are activation functions important in neural networks?

Follow-up Questions Interviewer Can Ask:

What happens if we remove activation functions?
Explain non-linearity in neural networks.
Compare ReLU, Sigmoid, and Tanh.
Why is ReLU commonly used in deep networks?
What problem does GELU solve?

Q54. Compare Activation Functions

Interviewer Asked:

Explain the difference between ReLU, Sigmoid, Tanh, and GELU activation functions.

Follow-up Questions Interviewer Can Ask:

Why does sigmoid suffer from vanishing gradients?
Why is GELU used in Transformers?
When would you use sigmoid in modern AI systems?
What happens when ReLU outputs zero for many neurons?

Q55. Explain Vanishing Gradient Problem

Interviewer Asked:

What is the vanishing gradient problem, and how do you solve it?

Follow-up Questions Interviewer Can Ask:

Why does it happen in deep networks?
Which activation functions reduce this problem?
How do residual connections help?
Why are ResNet architectures successful?
Does Transformer architecture suffer from this problem?

Q56. Explain CNN Architecture

Interviewer Asked:

Explain Convolutional Neural Network architecture and where it is used.

Follow-up Questions Interviewer Can Ask:

How does convolution operation work?
What are filters and kernels?
Why do CNNs work well for images?
Explain pooling layers.
CNN vs Vision Transformer — which would you choose?

Q57. Explain Convolution Operation

Interviewer Asked:

Explain how convolution works inside a CNN.

Follow-up Questions Interviewer Can Ask:

What is the purpose of filters?
How does stride affect output size?
What is padding and why is it used?
How are feature maps generated?

Q58. Explain RNN, LSTM, and GRU

Interviewer Asked:

Explain the difference between RNN, LSTM, and GRU architectures.

Follow-up Questions Interviewer Can Ask:

Why do traditional RNNs struggle with long sequences?
How does LSTM solve memory problems?
Explain forget gate, input gate, and output gate.
Why are Transformers preferred over LSTMs?

Q59. Why Did Transformers Replace RNNs?

Interviewer Asked:

Why did Transformers become more popular than RNNs for language models?

Follow-up Questions Interviewer Can Ask:

What limitation of RNNs does attention solve?
How does parallel processing work in Transformers?
Why can Transformers handle longer context?
Are RNNs completely obsolete?

Q60. Explain Transformer Architecture

Interviewer Asked:

Explain the complete Transformer architecture used in modern Large Language Models.

Follow-up Questions Interviewer Can Ask:

Explain encoder and decoder blocks.
What happens inside a Transformer layer?
Why do we stack multiple Transformer layers?
How does information flow through the model?
How does GPT architecture differ from the original Transformer?

Q61. Explain Self-Attention Mechanism

Interviewer Asked:

Explain self-attention and why it is important in Transformers.

Follow-up Questions Interviewer Can Ask:

How are attention scores calculated?
Explain Query, Key, and Value vectors.
Why do we divide by the square root of dimension?
How does attention understand relationships between words?

Q62. Explain Query, Key, and Value

Interviewer Asked:

What are Query, Key, and Value vectors in attention mechanism?

Follow-up Questions Interviewer Can Ask:

How are Q, K, and V generated?
What happens if the Key vector does not match the Query?
How does this help language understanding?
Explain attention calculation step-by-step.

Q63. Explain Multi-Head Attention

Interviewer Asked:

Why do Transformers use multi-head attention instead of a single attention layer?

Follow-up Questions Interviewer Can Ask:

What does each attention head learn?
How are multiple heads combined?
What happens if we increase the number of heads?
How does multi-head attention improve performance?

Q64. Explain Positional Encoding

Interviewer Asked:

Transformers process tokens in parallel. How does the model understand word order?

Follow-up Questions Interviewer Can Ask:

Why is positional information needed?
How is positional encoding calculated?
Learned vs fixed positional embeddings?
How does positional encoding affect long context?

Q65. Encoder vs Decoder Architecture

Interviewer Asked:

Explain the difference between encoder-only, decoder-only, and encoder-decoder Transformer models.

Follow-up Questions Interviewer Can Ask:

Where are encoder models used?
Why does GPT use decoder-only architecture?
Why does BERT use encoder-only architecture?
When would you use encoder-decoder models?

Q66. Why Does GPT Use Decoder-Only Architecture?

Interviewer Asked:

Why are GPT models based on decoder-only Transformers?

Follow-up Questions Interviewer Can Ask:

How does next-token prediction work?
What is causal masking?
Why can't GPT see future tokens during training?
How does this architecture generate text?

Q67. Explain Layer Normalization

Interviewer Asked:

What is Layer Normalization and why is it used in Transformers?

Follow-up Questions Interviewer Can Ask:

LayerNorm vs BatchNorm?
Why is BatchNorm difficult for NLP models?
Where is normalization applied in Transformer blocks?

Q68. Explain Dropout

Interviewer Asked:

How does dropout help neural networks generalize better?

Follow-up Questions Interviewer Can Ask:

What happens during training vs inference?
Why does dropout reduce overfitting?
Would you use dropout in LLM training?

Q69. Optimize Deep Learning Training

Interviewer Asked:

Your deep learning model takes days to train. How would you optimize the training process?

Follow-up Questions Interviewer Can Ask:

How does mixed precision training help?
How do batch size and learning rate affect training?
How do you optimize GPU utilization?
When would you use distributed training?

Q70. Explain Transfer Learning

Interviewer Asked:

Explain transfer learning and why it is widely used in AI.

Follow-up Questions Interviewer Can Ask:

When should you fine-tune a pretrained model?
Feature extraction vs fine-tuning?
Why does transfer learning reduce training cost?

Q71. Reduce GPU Memory Usage

Interviewer Asked:

Your LLM training is running out of GPU memory. How would you solve it?

Follow-up Questions Interviewer Can Ask:

Explain quantization.
Explain gradient checkpointing.
How does LoRA reduce memory usage?
What is mixed precision training?

Q72. Batch Normalization vs Layer Normalization

Interviewer Asked:

Compare Batch Normalization and Layer Normalization.

Follow-up Questions Interviewer Can Ask:

Why do Transformers prefer LayerNorm?
Where is BatchNorm commonly used?
How does normalization improve training stability?

Q73. Debug a Neural Network Not Learning

Interviewer Asked:

Your neural network training loss is not decreasing. How would you debug it?

Follow-up Questions Interviewer Can Ask:

How do you check learning rate problems?
How do you verify data quality?
How do you check gradients?
How do you identify architecture problems?

Q74. Choose Neural Network Architecture

Interviewer Asked:

How do you decide which neural network architecture to use for a problem?"

Follow-up Questions Interviewer Can Ask:

CNN vs Transformer — how do you decide?
How does dataset size influence architecture selection?
When would you use pretrained models?

Q75. Design a Deep Learning Solution

Interviewer Asked:

Design a deep learning solution for a real-world business problem.

Follow-up Questions Interviewer Can Ask:

How will you collect training data?
How will you select the model architecture?
How will you train and evaluate the model?
How will you deploy it?
How will you monitor performance?

SECTION 4: Generative AI & Large Language Model (LLM) Interview Questions

Q76. Explain Large Language Model Architecture

Interviewer Asked:

Explain the complete architecture of a Large Language Model like GPT.

Follow-up Questions Interviewer Can Ask:

What are the main components inside an LLM?
Explain the role of tokenizer, embeddings, and Transformer layers.
How does an LLM learn language patterns?
What happens during LLM pre-training?
How is GPT different from traditional NLP models?

Q77. Explain What Happens When a User Sends a Prompt to an LLM

Interviewer Asked:

Walk me through what happens internally when a user enters a prompt into ChatGPT.

Follow-up Questions Interviewer Can Ask:

How does tokenization happen?
How are tokens converted into embeddings?
What happens inside Transformer layers?
How does the model generate the next token?
Why does response generation become slower with longer conversations?

Q78. Explain Tokenization in LLMs

Interviewer Asked:

How does tokenization work in Large Language Models?

Follow-up Questions Interviewer Can Ask:

Why don't LLMs process complete words directly?

What is BPE tokenization?
How does token count affect API cost?
What happens when the model sees an unknown word?
How do different languages affect tokenization?

Q79. Explain Embeddings in Generative AI

Interviewer Asked:

What are embeddings, and why are they important in AI applications?

Follow-up Questions Interviewer Can Ask:

How are embeddings generated?
What information does an embedding vector represent?
How are embeddings used in RAG systems?
Why do similar meanings have similar vectors?
How do you select an embedding model?

Q80. Explain Context Window in LLMs

Interviewer Asked:

What is context window in Large Language Models?

Follow-up Questions Interviewer Can Ask:

What happens when input exceeds the context limit?
How does a larger context window affect cost?
Does a larger context window always improve performance?
How would you handle documents larger than the context window?

Q81. Explain Next Token Prediction

Interviewer Asked:

How does an LLM generate the next word or token?

Follow-up Questions Interviewer Can Ask:

Explain probability distribution over tokens.
How does temperature change output?
Explain Top-k and Top-p sampling.
Why can the same prompt generate different answers?

Q82. Temperature, Top-k, and Top-p Sampling

Interviewer Asked:

Explain different decoding strategies used during LLM generation.

Follow-up Questions Interviewer Can Ask:

When would you use low temperature?
When would you use high temperature?
Difference between Top-k and Top-p?
Which settings would you use for a coding assistant?

Q83. Explain LLM Hallucination

Interviewer Asked:

Why do Large Language Models hallucinate?

Follow-up Questions Interviewer Can Ask:

Is hallucination a model problem or data problem?
How would you reduce hallucination in an enterprise application?
How does RAG help reduce hallucinations?
How would you evaluate hallucination rate?

Q84. Prompt Engineering Techniques

Interviewer Asked:

What prompt engineering techniques have you used to improve LLM responses?

Follow-up Questions Interviewer Can Ask:

Explain zero-shot prompting.
Explain few-shot prompting.
What is role prompting?
How do you create effective system prompts?
How do you debug a bad prompt?

Q85. Chain-of-Thought Prompting

Interviewer Asked:

Explain chain-of-thought prompting and when you would use it.

Follow-up Questions Interviewer Can Ask:

Why does reasoning improve with structured prompts?
What are limitations of chain-of-thought?
How do reasoning models differ from normal LLMs?
How would you evaluate reasoning ability?

Q86. Fine-Tuning vs Prompt Engineering

Interviewer Asked:

When would you use fine-tuning instead of prompt engineering?

Follow-up Questions Interviewer Can Ask:

What problems can fine-tuning solve?
What problems should not use fine-tuning?
How much data is required for fine-tuning?
What are the risks of fine-tuning?

Q87. Fine-Tuning vs RAG

Interviewer Asked:

Your company wants an AI assistant trained on internal documents. Would you choose fine-tuning or RAG?

Follow-up Questions Interviewer Can Ask:

Explain the difference between RAG and fine-tuning.
When is RAG the better choice?
Can you combine RAG and fine-tuning?
How do you maintain updated knowledge?

Q88. Explain Instruction Tuning

Interviewer Asked:

How are LLMs trained to follow human instructions?

Follow-up Questions Interviewer Can Ask:

What happens during supervised fine-tuning?
How does instruction tuning improve usability?
How is instruction tuning different from pre-training?

Q89. Explain RLHF

Interviewer Asked:

Explain Reinforcement Learning from Human Feedback (RLHF).

Follow-up Questions Interviewer Can Ask:

Why is RLHF required after pre-training?
Explain reward model training.
What are limitations of RLHF?
What alternatives exist to RLHF?

Q90. Explain DPO (Direct Preference Optimization)

Interviewer Asked:

What is DPO and how is it different from RLHF?

Follow-up Questions Interviewer Can Ask:

Why was DPO introduced?
What are the advantages of DPO?
Compare DPO, PPO, and RLHF.

Q91. Explain LoRA Fine-Tuning

Interviewer Asked:

Explain LoRA and why it is widely used for LLM fine-tuning.

Follow-up Questions Interviewer Can Ask:

How does LoRA reduce trainable parameters?
LoRA vs full fine-tuning?
How does LoRA reduce GPU memory requirements?
What are LoRA adapters?

Q92. Explain QLoRA

Interviewer Asked:

What is QLoRA and how does it make LLM fine-tuning cheaper?

Follow-up Questions Interviewer Can Ask:

How does quantization work?
Difference between LoRA and QLoRA?
What are the tradeoffs of quantization?

Q93. Reduce LLM Inference Cost

Interviewer Asked:

Your company is spending thousands of dollars monthly on LLM APIs. How would you reduce the cost?

Follow-up Questions Interviewer Can Ask:

How does prompt optimization reduce cost?
How does caching help?
When would you use smaller models?
How does quantization reduce cost?

Q94. Deploy an LLM Application in Production

Interviewer Asked:

Explain how you would deploy a Generative AI application into production.

Follow-up Questions Interviewer Can Ask:

How will you manage model versions?
How will you handle scaling?
How will you monitor quality?
How will you secure user data?

Q95. Evaluate LLM Performance

Interviewer Asked:

How do you evaluate the performance of a Large Language Model application?

Follow-up Questions Interviewer Can Ask:

Which metrics would you measure?
How do you evaluate answer quality?
How do you measure hallucination?
How do you perform human evaluation?

Q96. Build an Enterprise AI Assistant

Interviewer Asked:

Design an AI assistant for employees that can answer company-related questions.

Follow-up Questions Interviewer Can Ask:

How will you connect company documents?
How will you protect sensitive information?
How will you handle user permissions?
How will you evaluate accuracy?

Q97. Explain AI Guardrails

Interviewer Asked:

How would you make an LLM application safe for enterprise users?

Follow-up Questions Interviewer Can Ask:

What types of risks do LLM applications have?
How do you prevent harmful responses?
How do you implement content filtering?
How do you monitor AI safety?

Q98. Multi-Modal AI Models

Interviewer Asked:

Explain how modern AI models handle text, images, and audio together.

Follow-up Questions Interviewer Can Ask:

How are different modalities combined?
What are vision-language models?

How would you design a multimodal AI application?

Q99. Open Source LLM vs Closed Source LLM

Interviewer Asked:

Compare open-source LLMs and proprietary models. Which would you choose for enterprise use?

Follow-up Questions Interviewer Can Ask:

What are advantages of open-source models?
What are advantages of commercial APIs?
How do you choose a model for production?

Q100. Design a Generative AI Product

Interviewer Asked:

Design a complete Generative AI product from idea to production.

Follow-up Questions Interviewer Can Ask:

How would you select the right model?
How would you design the architecture?
How would you handle scalability?
How would you measure product success?
How would you reduce operational cost?

SECTION 5: RAG, Vector Database & AI Search Interview Questions

Q101. Design a Production-Ready RAG System

Interviewer Asked:

Design a Retrieval Augmented Generation (RAG) system for an enterprise that allows employees to search internal company documents using natural language.

Follow-up Questions Interviewer Can Ask:

Explain the complete architecture of your RAG system.
How do documents move from storage to final LLM response?
How would you select chunk size and chunking strategy?
Which embedding model would you choose and why?
How would you evaluate RAG system accuracy?
How would you handle millions of documents?

Q102. Explain RAG Architecture

Interviewer Asked:

Explain how Retrieval Augmented Generation works internally.

Follow-up Questions Interviewer Can Ask:

Why was RAG introduced?
What problem does RAG solve in LLM applications?
Explain retrieval phase and generation phase.
Where can RAG fail?
How is RAG different from traditional search?

Q103. Document Chunking Strategy in RAG

Interviewer Asked:

How would you design a document chunking strategy for a RAG application?

Follow-up Questions Interviewer Can Ask:

Q101. Design a Production-Ready RAG System

Interviewer Asked:

Design a Retrieval Augmented Generation (RAG) system for an enterprise that allows employees to search internal company documents using natural language.

Follow-up Questions Interviewer Can Ask:

Explain the complete architecture of your RAG system.
How do documents move from storage to final LLM response?
How would you select chunk size and chunking strategy?
Which embedding model would you choose and why?
How would you evaluate RAG system accuracy?
How would you handle millions of documents?

Q102. Explain RAG Architecture

Interviewer Asked:

Explain how Retrieval Augmented Generation works internally.

Follow-up Questions Interviewer Can Ask:

Why was RAG introduced?
What problem does RAG solve in LLM applications?
Explain retrieval phase and generation phase.
Where can RAG fail?
How is RAG different from traditional search?

Q103. Document Chunking Strategy in RAG

Interviewer Asked:

How would you design a document chunking strategy for a RAG application?

Follow-up Questions Interviewer Can Ask:

What chunk size would you choose and why?
Explain fixed-size chunking.
Explain semantic chunking.
How does chunk overlap improve retrieval?
How do you handle tables, PDFs, and images?

Q104. Improve RAG Retrieval Accuracy

Interviewer Asked:

Your RAG application retrieves irrelevant documents. How would you improve retrieval quality?

Follow-up Questions Interviewer Can Ask:

Would you change embeddings or retrieval strategy?
Explain hybrid search.
What is re-ranking?
How does metadata filtering improve results?
How do you debug retrieval failures?

Q105. Explain Vector Database

Interviewer Asked:

What is a vector database and why is it required for GenAI applications?

Follow-up Questions Interviewer Can Ask:

How are embeddings stored inside a vector database?
How does vector search work internally?
Difference between SQL database and vector database?
When would you use both together?
How does vector indexing improve performance?

Q106. Explain Similarity Search

Interviewer Asked:

How does a vector database find similar documents?

Follow-up Questions Interviewer Can Ask:

Explain nearest neighbor search.
Cosine similarity vs Euclidean distance?
Why is exact search expensive for billions of vectors?
How does approximate nearest neighbor search work?

Q107. Explain HNSW Algorithm

Interviewer Asked:

Explain how HNSW indexing works in vector databases.

Follow-up Questions Interviewer Can Ask:

Why is HNSW faster than brute-force search?
What are graph-based indexes?
How does HNSW balance speed and accuracy?
What parameters affect HNSW performance?

Q108. FAISS vs Pinecone vs Other Vector Databases

Interviewer Asked:

Compare FAISS and managed vector databases like Pinecone.

Follow-up Questions Interviewer Can Ask:

When would you choose FAISS?
When would you use a managed vector database?
How do you scale vector search?
What factors influence your selection?

Q109. Build a Semantic Search Engine

Interviewer Asked:

Design a semantic search engine using embeddings and LLMs.

Follow-up Questions Interviewer Can Ask:

How is semantic search different from keyword search?
How would you rank search results?
How would you handle millions of queries?
How would you improve search relevance?

Q110. Hybrid Search Architecture

Interviewer Asked:

Explain hybrid search and why companies use it in RAG systems.

Follow-up Questions Interviewer Can Ask:

Difference between keyword search and vector search?
What is BM25?
How do you combine lexical and semantic search?
When does hybrid search perform better?

Q111. RAG Evaluation

Interviewer Asked:

How would you evaluate the performance of a RAG application?

Follow-up Questions Interviewer Can Ask:

How do you measure retrieval quality?
How do you measure answer quality?
What metrics would you track?
How do you measure hallucination?
How do you create a RAG evaluation dataset?

Q112. Reduce RAG Hallucination

Interviewer Asked:

Your RAG chatbot provides incorrect answers even though documents exist. How will you fix it?

Follow-up Questions Interviewer Can Ask:

Is the issue retrieval or generation?
How would you debug the pipeline?
Would you modify prompts or embeddings?
How would you add confidence scoring?

Q113. Build Multi-Document RAG System

Interviewer Asked:

Design a RAG system that can answer questions from thousands of different documents.

Follow-up Questions Interviewer Can Ask:

How will you organize documents?
How will you apply metadata filtering?
How will you handle conflicting information?
How will you update documents?

Q114. Real-Time RAG System Design

Interviewer Asked:

Design a real-time RAG system where documents are updated continuously.

Follow-upQuestions Interviewer Can Ask:

How will you update embeddings?
How will you handle deleted documents?
How will you maintain consistency?
How will you reduce update latency?

Q115. Secure Enterprise RAG

Interviewer Asked:

How would you design a secure RAG system for confidential company data?

Follow-up Questions Interviewer Can Ask:

How will you implement access control?
How will you prevent data leakage?
How will you handle different user permissions?
How will you audit AI responses?

Q116. RAG vs Fine-Tuning

Interviewer Asked:

Your company wants an AI assistant trained on internal knowledge. Would you choose RAG or fine-tuning?

Follow-up Questions Interviewer Can Ask:

What are the advantages of RAG?

When is fine-tuning better?
Can both approaches be combined?
Which approach is easier to maintain?

Q117. Optimize RAG Cost

Interviewer Asked:

Your RAG application is expensive to operate. How would you reduce cost?

Follow-up Questions Interviewer Can Ask:

How would you reduce token usage?
How would you optimize embeddings?
How would you select smaller models?
How would caching help?

Q118. RAG Pipeline Debugging

Interviewer Asked:

Your RAG system gives poor answers. Explain your debugging approach.

Follow-up Questions Interviewer Can Ask:

How will you check document quality?
How will you test retrieval results?
How will you evaluate prompts?
How will you identify bottlenecks?

Q119. Scale RAG for Millions of Users

Interviewer Asked:

Design a RAG system that supports millions of users.

Follow-up Questions Interviewer Can Ask:

How will you scale vector search?
How will you handle concurrent requests?
How will you reduce latency?
How will you manage infrastructure cost?

Q120. Design an Enterprise Knowledge Assistant

Interviewer Asked:

Design an AI knowledge assistant like ChatGPT for an organization.

Follow-up Questions Interviewer Can Ask:

How will you connect enterprise data sources?
How will you manage user authentication?
How will you ensure accurate answers?
How will you monitor quality?
How will you make the system production-ready?

SECTION 6: Agentic AI & Multi-Agent System Interview Questions

Q121. Explain Agentic AI Architecture

Interviewer Asked:

What is Agentic AI, and how is it different from a traditional chatbot?

Follow-up Questions Interviewer Can Ask:

What are the main components of an AI agent?
Explain the role of LLM, memory, tools, and planning in an agent.
Why are AI agents becoming popular now?
What limitations do normal chatbots have compared to agents?
Give examples of real-world AI agent applications.

Q122. Design an Autonomous AI Agent

Interviewer Asked:

Design an AI agent that can research a topic, collect information, and generate a final report.

Follow-up Questions Interviewer Can Ask:

How will the agent decide what actions to take?
How will you implement planning?
How will the agent use external tools?
How will you prevent incorrect actions?
How will you evaluate agent performance?

Q123. AI Agent vs Chatbot

Interviewer Asked:

Explain the difference between an AI chatbot and an autonomous AI agent.

Follow-up Questions Interviewer Can Ask:

Does every chatbot qualify as an agent?

"How would you deploy and update agents?"

What makes an agent autonomous?
How does an agent maintain state and memory?
When would you choose a chatbot instead of an agent?

Q124. Explain ReAct Framework

Interviewer Asked:

Explain the ReAct framework used in AI agents.

Follow-up Questions Interviewer Can Ask:

What does Reason + Act mean in ReAct?
How does the agent decide when to use tools?
How is ReAct different from simple prompting?
What are limitations of ReAct agents?

Q125. Tool Calling in AI Agents

Interviewer Asked:

How do AI agents interact with external tools and APIs?

Follow-up Questions Interviewer Can Ask:

Explain function calling in LLMs.
How does an agent select the correct tool?
How do you validate tool inputs?
How do you handle tool execution failures?
How do you secure external tool access?

Q126. Memory Management in AI Agents

Interviewer Asked:

How would you design memory for an AI agent?

Follow-up Questions Interviewer Can Ask:

Explain short-term vs long-term memory.
Where would you store agent memories?
How do you retrieve relevant memories?
How do you prevent memory overload?
How does memory improve agent performance?

Q127. Multi-Agent System Architecture

Interviewer Asked:

Design a multi-agent AI system where multiple agents collaborate to complete a complex task.

Follow-up Questions Interviewer Can Ask:

How do agents communicate with each other?
How do you assign roles to different agents?
What happens if agents disagree?
How do you coordinate agent execution?
What are the challenges of multi-agent systems?

Q128. Agent Orchestration

Interviewer Asked:

How would you orchestrate multiple AI agents in a production environment?

Follow-up Questions Interviewer Can Ask:

Would you use a central controller or decentralized agents?
How do you manage agent workflows?
How do you handle failed agent tasks?
How do you monitor agent execution?

Q129. AI Agent Planning

Interviewer Asked:

How does an AI agent plan and break down a complex goal into smaller tasks?

Follow-up Questions Interviewer Can Ask:

Explain task decomposition.
What planning strategies can agents use?
How does the agent decide task priority?
How do you prevent infinite loops?

"How would you deploy and update agents?"

Q130. Self-Reflection and Self-Correction Agents

Interviewer Asked:

Explain how self-correcting AI agents work.

Follow-up Questions Interviewer Can Ask:

Why do agents need reflection?
How does an agent evaluate its own output?
What are reflection loops?
How do you prevent unnecessary iterations?

Q131. Build an AI Coding Agent

Interviewer Asked:

Design an AI coding assistant that can understand requirements, write code, test, and fix bugs.

Follow-up Questions Interviewer Can Ask:

How will the agent understand a large codebase?
How will you provide code context?
How will the agent execute and test code safely?
How will you prevent generating insecure code?
How would you evaluate coding accuracy?

Q132. Build an AI Research Agent

Interviewer Asked:

Design an AI research assistant that can search information and create reports.

Follow-up Questions Interviewer Can Ask:

Which tools would the agent need?
How will you verify information accuracy?
How will you handle conflicting sources?
How will you store research history?

Q133. Agent Evaluation

Interviewer Asked:

How would you evaluate the performance of an AI agent?

Follow-up Questions Interviewer Can Ask:

What metrics would you measure?
How do you evaluate task completion?
How do you measure agent reliability?
How do you test agents before production?
How do you evaluate tool usage accuracy?

Q134. AI Agent Safety

Interviewer Asked:

What are the major safety challenges with autonomous AI agents?

Follow-up Questions Interviewer Can Ask:

How do you prevent unwanted actions?
How do you implement human approval steps?
How do you control agent permissions?
How do you audit agent decisions?

Q135. Design an Enterprise Multi-Agent Platform

Interviewer Asked:

Design a scalable enterprise multi-agent AI platform.

Follow-up Questions Interviewer Can Ask:

How would you manage thousands of agents?
How would agents share information securely?
How would you monitor agent behavior?
How would you manage costs?

SECTION 7: AI System Design, LLMOps & Production AI Interview Questions

Q136. Design ChatGPT at Enterprise Scale

Interviewer Asked:

Design a ChatGPT-like AI application that can support millions of users.

Follow-up Questions Interviewer Can Ask:

Explain the complete high-level architecture.
How will you handle millions of simultaneous users?
How will you manage LLM inference requests?
How will you reduce GPU infrastructure cost?
How will you maintain conversation history?
How will you monitor system performance?

Q137. Design an AI Coding Assistant Like GitHub Copilot

Interviewer Asked:

Design an AI coding assistant that can generate code suggestions inside an IDE.

Follow-up Questions Interviewer Can Ask:

How will you provide code context to the LLM?
How will you handle large codebases?
Would you use RAG, fine-tuning, or both?
How will you evaluate generated code quality?
How will you prevent insecure code generation?

Q138. Design an AI Customer Support Agent

Interviewer Asked:

Design an AI customer support system that can handle millions of customer queries.

Follow-up Questions Interviewer Can Ask:

How will the AI access customer information?
How will you connect knowledge bases?
How will you handle unknown questions?
When should the AI transfer to a human agent?
How will you measure customer satisfaction?

Q139. Design a Real-Time AI Search Engine

Interviewer Asked:

Design an AI-powered search engine that understands user intent.

Follow-up Questions Interviewer Can Ask:

Traditional search vs AI search — what is the difference?
How will you use embeddings?
How will you rank search results?
How will you handle billions of documents?
How will you reduce search latency?

Q140. Design an AI Recommendation System

Interviewer Asked:

Design a recommendation engine like Netflix, YouTube, or Amazon.

Follow-up Questions Interviewer Can Ask:

What data will you collect from users?
How will you generate recommendations in real time?
How will you solve cold-start problems?
How will you evaluate recommendation quality?
How will you avoid recommendation bias?

Q141. Design a Large-Scale RAG Platform

Interviewer Asked:

Design a RAG platform that supports millions of enterprise documents and users.

Follow-up Questions Interviewer Can Ask:

How will you store and index documents?
How will you manage embeddings at scale?
How will you handle frequent document updates?
How will you optimize retrieval speed?
How will you ensure data security?

Q142. Deploy LLM Application in Production

Interviewer Asked:

Explain how you would deploy an LLM application from development to production.

Follow-up Questions Interviewer Can Ask:

What infrastructure components are required?
How will you manage different model versions?
How will you perform testing before release?
How will you handle deployment failures?
How will you perform rollback?

Q143. Explain LLMOps Architecture

Interviewer Asked:

What is LLMOps and why is it important for production AI systems?

Follow-up Questions Interviewer Can Ask:

How is LLMOps different from MLOps?
What components are included in LLMOps?
How do you monitor LLM applications?
How do you manage prompts and versions?
How do you track model performance?

Q144. Monitor Generative AI Applications

Interviewer Asked:

How would you monitor the quality and performance of a Generative AI application?

Follow-up Questions Interviewer Can Ask:

What metrics would you monitor?
How would you detect hallucinations?
How would you monitor latency?
How would you monitor token usage?
How would you identify bad responses?

Q145. Reduce LLM Inference Latency

Interviewer Asked:

Your AI application response time is too slow. How would you optimize LLM inference?

Follow-up Questions Interviewer Can Ask:

What factors impact LLM latency?
How does KV cache improve performance?
How does quantization reduce latency?
How does model size affect inference speed?
When would you use a smaller model?

Q146. Reduce LLM Application Cost

Interviewer Asked:

Your company's GenAI application cost is increasing rapidly. How would you optimize expenses?

Follow-up Questions Interviewer Can Ask:

How does prompt optimization reduce cost?
How can caching reduce API calls?
When would you choose open-source models?
How does batching improve cost efficiency?
How would you calculate cost per user?

Q147. Design AI Model Security Architecture

Interviewer Asked:

How would you secure an enterprise AI application?

Follow-up Questions Interviewer Can Ask:

How do you protect sensitive user data?
How do you prevent prompt injection attacks?
How do you secure RAG systems?
How do you control model access?
How do you audit AI decisions?

Q148. Design AI Evaluation Framework

Interviewer Asked:

How would you build an evaluation framework for an AI application?

Follow-up Questions Interviewer Can Ask:

How do you measure accuracy?
How do you evaluate LLM responses?
How do you create test datasets?
How do you perform human evaluation?
How do you compare different models?

Q149. Design a Complete Enterprise Generative AI Platform

Interviewer Asked:

Design an enterprise Generative AI platform from data ingestion to user interaction.

Follow-up Questions Interviewer Can Ask:

How will you manage enterprise data?
How will you select foundation models?
How will you implement RAG and agents?
How will you deploy and scale the platform?
How will you ensure security and compliance?

Q150. Design an AI Platform for the Future

Interviewer Asked:

If you had to build an AI platform capable of supporting future AI applications, how would you design it?

Follow-up Questions Interviewer Can Ask:

"What architecture would you choose?"
"How would you support multiple AI models?"
"How would you integrate AI agents?"
"How would you manage data, models, and users?"
"What challenges do you expect at billion-user scale?"



Thank You!